Figure 4: Use-Case Diagram

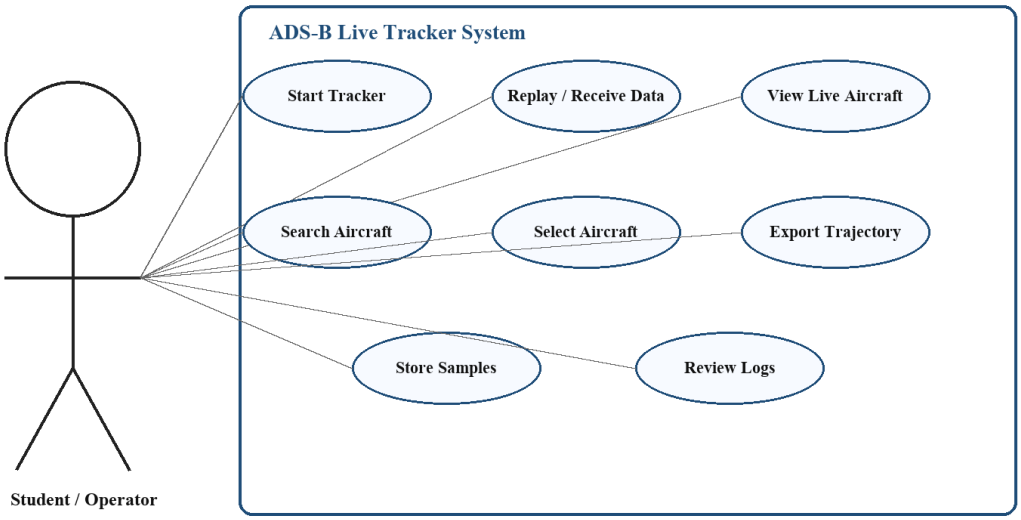


Figure 4: Use-Case Diagram

3. REQUIREMENT SPECIFICATION

3.1 System Inputs

The system accepts PingStation data in two modes. In simulation mode, captured hex log data is replayed from a local file so that the tracker can be tested without live receiver hardware. In live mode, the receiver listens on a configured host and port and decodes incoming stream data. Environment variables are used to switch modes and configure paths, ports and logging behavior.

3.2 System Outputs

- Live aircraft JSON returned by the /api/aircraft endpoint.
- Latest matrix-style aircraft records from /api/training/matrix.
- Trajectory history through /api/training/trajectory and CSV export through /api/training/trajectory/export.
- Daily trajectory CSV files under logs/training_csv.
- ICAO-wise Excel workbooks and a master ICAO workbook.
- Dashboard visualization using map markers, aircraft list, route panel and altitude chart.